

Reader Supplement to
*Generic Identifiability and Directed Containment for Strongly
Tree-Child Level-2 Networks under the Kimura Two-Parameter
Model:*
The Principal Positive Domain and Strict Continuous Time

Alec Kriebel
Independent Researcher
ORCID: [0009-0001-9320-500X](https://orcid.org/0009-0001-9320-500X)
Corresponding email: me@aleckriebel.com

August 2026

Abstract

This supplement is a reader-oriented map from the mathematical proof to the exact finite certificate package. It separates hand proofs from the bounded computer-assisted residue, defines the primitive completion grammar and raw relation object, records all censuses and restoration/probe invariants, gives representative certificate schemas and the complete weak-class witness, and lists the minimal artifacts and commands needed to replay each theorem layer. The finite principal-domain theorem and classification universe remain unchanged; the qualification-evidence lock is resealed when verifier-facing tests or their provenance bindings are repaired. The article and this supplement are separately versioned submission sources.

Contents

1	How to read the proof	2
2	Convention crosswalk	3
3	Primitive graph grammar	4
3.1	Core segments and minimum repairs	4
3.2	Completion grammar and the numbers 831, 1983, and 4155	4
3.3	Raw relation schema	4
4	Certificate decision tree	5
5	Finite census and current compression boundary	6
5.1	Authoritative raw partitions	6
5.2	Audited proof-compression table	6

5.3	Restoration forests	7
6	Coherent probes and arbitrary subdivision words	7
7	Representative exact calculations	8
7.1	Tree–sunlet factorization	8
7.2	Bridge normalizer	9
7.3	Certificate schema for a direct polynomial	9
8	Printed exact-certificate appendix	9
8.1	Coordinate dictionaries	9
8.2	The 23 literal quadratic bodies	10
8.3	Five high-degree bases and their licensed transports	11
8.4	Three worked paths	13
8.5	Provenance and replay	14
9	Full weak-class certificate	15
10	Theorem-to-artifact crosswalk	16
10.1	Printed finite-layer replay metadata	19
11	Frozen hash anchors	21
12	Replay protocol	21
13	Mutation coverage	22
14	Recent-source verification note	23
15	Submission metadata and licenses	23

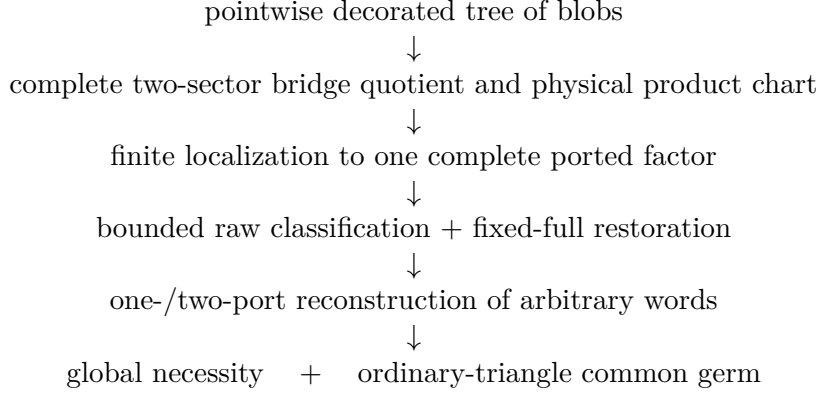
1 How to read the proof

The main article proves a global theorem from three qualitatively different types of input.

- P1. Analytic and algebraic proofs.** These include the physical K2P domain, subdivision and rerooting, pointwise quartet and whole-map $\mathcal{T}_i$ signs, the two-sector bridge fibre, physical local products, marginal submersions, localization, genericity, continuous-time gluing, and the weak-class construction.
- P2. Topology-only finite reduction.** The cycle/theta primitive theorem and minimum-repair grammar are graph-theoretic. They reduce an arbitrary strong level-2 factor to a finite rigid support plus ordered labelled words on finitely many directed segments.
- P3. Exact bounded computation.** Raw four- and five-port directions, rank exclusions, direct polynomial separators, restoration children, and coherent probes are exhausted by exact graph

and rational/symbolic arithmetic. This layer has independent regeneration and mutations, but is not presented as a complete short hand classification.

The logical dependency is



The quartet and $\mathcal{T}_i$ inequalities are pointwise. The polynomial and rank certificates are generic non-containment statements. Restoration applies only inside one assumed fixed full relation. Keeping these quantifiers separate is essential.

The independent replay boundary is also explicit. The package regenerates the primitive universes, checks exact graph relations and symbolic certificates, and subjects complete ledgers to verifier-facing mutations. It does not claim an additional all-family orbit partition built without either the submitted primitive atlas or canonicalizer, nor a second symbolic engine that independently re-expands every higher-degree polynomial body. Those would be stronger independent research audits, not premises of the theorem or of the finite replay reported here.

2 Convention crosswalk

All graphs use the following fixed convention.

- Rooted presentations are binary directed acyclic graphs; the root is the lowest stable ancestor.
- Only arcs entering reticulations retain arrowheads.
- The root is deleted once and its two incident arcs merge into one mixed edge. No later parallel-edge or degree-two cleanup is part of formation of the full standard semi-directed graph.
- Restriction/display operations may subsequently prune dead ends and suppress unlabelled bivalent vertices; they do not change the admissible-rooting universe of the original graph.
- “Strongly tree-child” quantifies over every admissible rooting; “weakly tree-child” requires at least one tree-child rooting.
- Isomorphisms preserve labels, ordinary edges, arrowheads, and vertex roles. Ordinary triangle equivalence forgets only the reticulation choice on one three-cycle and keeps its three external arms fixed.

The K2P character order is $(0, C, G, T)$, with equal sector $\{C, T\}$ and singleton sector $\{G\}$. Every edge has Fourier vector $(1, s, g, s)$ in

$$\mathcal{D}_+ = \{0 < s < 1, 0 < g < 1, g > 2s - 1\}.$$

The strict continuous-time subdomain is

$$\mathcal{D}_{\text{CT}} = \{0 < s < 1, s^2 < g < 1\}.$$

No mixed-sign stochastic component is part of the theorem.

3 Primitive graph grammar

3.1 Core segments and minimum repairs

After removing external port arms and suppressing ordinary non-event degree-two vertices, a nontrivial strong level-2 blob is a cycle or one of four directed theta event placements. Segment indices in the last column below refer to the left-to-right segment order.

core	directed segments	minimum repair sets
cycle	$S \rightarrow X, S \rightarrow X$	$\{0\}, \{1\}$
θ_0	$S \rightarrow U, S \rightarrow V, U \rightarrow X, V \rightarrow X, U \rightarrow V$	$\{2, 3\}, \{3, 4\}$
θ_1	$S \rightarrow U, S \rightarrow X, V \rightarrow X, U \rightarrow V, U \rightarrow V$	$\{2, 3\}, \{2, 4\}$
θ_2	$S \rightarrow U, S \rightarrow V, U \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1$	$\{2, 3\}, \{2, 5\}, \{3, 4\}, \{4, 5\}$
θ_3	$S \rightarrow U, S \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1, U \rightarrow V$	$\{2\}, \{4\}$

An occupied segment contains at least one port-bearing subdivision. The minimum repairs are the minimal occupancy sets satisfying the no-omnian and simple-graph requirements. A complete factor additionally retains one child role below every path-sink reticulation. Here “occupied” includes a named zero-character completion role inserted by the compiler; it need not be a selected physical port. Arbitrary physical factors arise by replacing each selected segment word by an ordered word of distinct labelled ports.

3.2 Completion grammar and the numbers 831, 1983, and 4155

For each core the compiler enumerates:

- (i) every allowed subset of path-sink child roles;
- (ii) every weak composition of the remaining labelled port roles among the directed segments;
- (iii) for a theta target, every minimum-repair tag; for a cycle target, the single untagged target convention (the two repaired cycle supports occur on the source side);
- (iv) the selected-incoming mode, in which a real incoming boundary remains a port, and the marginalized-incoming mode, in which that role has been omitted; and
- (v) every physical port permutation.

The directed completion grammar yields 831 four-port target descriptors in selected-incoming mode and 1,983 in marginalized-incoming mode. Five-port targets number 1,983 and 4,155 in the corresponding modes. These are repair-tagged descriptors, not necessarily distinct untagged mixed graphs. The compiler first weakly distributes the selected physical labels. For a chosen theta repair it then inserts one deterministically named zero-character leaf on each required segment whose physical word is empty, and inserts nothing extra on an already occupied required segment. The repair tag stays in the raw identity even when another repair witness yields the same final mixed graph. Unselected sink roles and a marginalized incoming role receive their own deterministic dummies. Consequently these roles consume no selected labels and the stars-and-bars count in the article is unchanged. The numbers are taken before source multiplication and port permutations. Thus

$$6(831 + 1983)4! = 405,216, \quad 4(1983 + 4155)5! = 2,946,240.$$

3.3 Raw relation schema

A raw directional relation is not an unlabelled topology hash. Its immutable identity includes:

- source primitive and exact repaired support;
- target primitive, completion descriptor, and selected graph;
- source and target incoming roles;
- physical port permutation and port match;
- omitted source and target roles;
- source-to-target direction;
- boundary incidence transport; and
- exact graph fingerprints.

Only after every raw ID is generated exactly once are presentations grouped into canonical terminal or restoration classes.

4 Certificate decision tree

The classifier uses proof types in a fixed precedence order.

type	mathematical implication	exact stored evidence
Displayed quartet	physical images are disjoint pointwise	distinct complete switching-split sets, selected witness split, and the appropriate leaf-permuted F or G polynomial.
Whole-map $\mathcal{T}_i$	physical images are disjoint pointwise	coordinate triple and orientation, exact whole-map pullbacks, and a strict Bernstein/sign certificate for the factorized pullback.
Directed rank	no full-dimensional source germ enters the target	nonzero exact source minor, target lower minor, symbolic target rank upper certificate, and strict source-rank $>$ target-rank comparison.
Polynomial separator	generic source-to-target containment is impossible	sparse integer polynomial $P(q)$, coefficientwise proof $P \circ \Phi_T = 0$, nonzero source pullback, and strict rational source witness.
Restoration parent	deferred fixed-full obligation	parent relation, each actual physical child attachment, exact parent-child transport restriction, and a terminal certificate at every leaf.
Isomorphism	physical parameter transport exists	exact labelled mixed-incidence isomorphism and parameter/port transport.
Ordinary triangle	one common full-dimensional regular germ exists	stored triangle edge set and coherent transport, combined with the exact rank-nine analytic construction in the article.

The target rank upper certificate is symbolic; a one-point Jacobian rank is never used as an upper bound. Polynomial witnesses are checked on the graph- generated K2P maps rather than accepted by a topology identifier.

5 Finite census and current compression boundary

5.1 Authoritative raw partitions

category	four-port theta	five-port θ_2
displayed quartet	360,408	2,942,592
whole-map $\mathcal{T}_i$	16,974	2,528
directed exact rank	23,822	800
direct terminal / quadratic	1,472	240
restoration member / isomorphism	2,540	80
total	405,216	2,946,240

The 1,472 four-port terminal presentations form 934 classes:

839 quadratics + 36 direct polynomials + 4 hard bindings + 20 isomorphisms + 35 triangles.

The 36 direct polynomials have three certified source-family tags: 22 θ_0 quintic port-orbit obstructions, 12 lower-theta quartics, and two θ_3 cubics. The 839 quadratic classes have eight literal certificate bodies. These repetitions make further theorem-template compression plausible, but equality of certificate bodies is not by itself a proof that all graph transports belong to one legal orbit.

5.2 Audited proof-compression table

The bounded compression pass has status **PC-PARTIAL**. It changes no theorem statement and discards no exact ledger. In particular, literal equality of polynomial bodies is not treated as a graph orbit, and the 297 restoration fingerprints are descriptive archetypes rather than a proved transport quotient.

layer	frozen units	compact units	exact residue retained
four-port quadratics	839	8	every coordinate transport
five-port θ_2 quadratics	96	4	every coordinate transport
three-port cycle quadratics	54	6	every coordinate transport
restoration quadratics	6	5	every coordinate transport
direct-36 high degree	36	27	all directional bindings; 3 certified proposition families, 5 stored base formulas
rank upper certificates	4,379	3,515+75	universal mechanism plus 75 exceptional representatives covering 864 descriptors
restoration parents	997	297	997 assignments, 2,540 roots, 36,824 edges, and 16 algebra-transport classes

The direct-36 row must not be read as three literal polynomials. The quintic, quartic, and cubic proposition families have 27 direction-specific bodies across the 36 records. The quadratic rows group only literal coordinate bodies inside their original coordinate convention or transports already certified by a frozen orbit package.

layer	uniform statement	load-bearing finite premise
primitive completions	one binomial-sum formula for all selected/marginalized cases	all labelled directed completion records; totals 405,216, 2,946,240, and 13,440
restoration	four terminal mechanisms and one depth-two continuation mechanism	the exact 997-parent assignment and every physical parent-child transport
probe reconstruction	one-/two-port coherence implies arbitrary attachment-word recovery	176 anchors, 29,964 one-port rows, 544,571 two-port rows, 67,741 transports, and 4,379 restrictions

The machine-readable theorem-to-artifact crosswalk is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`. It records, for each theorem layer, exact file hashes and declared schemas, producer, replay, mutation, and environment bindings. The original frozen principal-domain computational-evidence lock does not itself contain an end-to-end timing record. For C01–C10 and C12–C13, the theorem-layer crosswalk leaves end-to-end runtimes null, preserving locked component observations without summing them. Separately, its global C11 entry and the submission bundle bind a detached clean-checkout replay of the qualified lock. The exact candidate commit, layer count, wall and internal times, memory measurements, report hash, and telemetry hash are recorded only in the externally bound replay artifacts and generated theorem-to-artifact crosswalk; they are deliberately not compiled into this source. No byte-bound end-to-end quick-suite runtime is claimed.

5.3 Restoration forests

The corrected four-port restoration forest has 997 canonical parents and 2,540 physical member roots. Its first layer is

quartet separator	35,758	
whole-map $\mathcal{T}_i$	606	
quadratic	148	(36,568 first children).
transported quartic	24	
continuation	32	

The 32 continuations have 256 second children: 248 quartet and eight $\mathcal{T}_i$. Hence there are 36,824 edges, 36,792 exact separator leaves, maximum depth two, no cycle, no missing child, and no unresolved record. Every child transport restricts to its parent's boundary transport.

The five-port θ_2 dummy forest has 56 roots and 864 one-parent descendants. Its 832 leaves are 760 quartet exclusions and 72 labelled isomorphisms. The cycle layer has 5,964 restoration roots and 536,364 physical completions, of which 535,920 are quartet-separated, 300 are $\mathcal{T}_i$ -separated, 132 have exact quadratics, and 12 are isomorphic.

The logical implication is always fixed-full. An assumed full relation fixes the omitted physical label, source edge, target edge, role, and transport. That actual completion is one child in the forest. Source marginal openness then turns its separator into a contradiction. No abstract parent relation is lifted, and no target deletion map is inverted.

6 Coherent probes and arbitrary subdivision words

The equality-anchor input contains 176 physical relations:

origin	ports	records
four-port direct terminals	4	26
restored four-port terminals	5	17
θ_2 direct anchors	5	24
θ_2 one-role restorations	6	40
θ_2 two-role paths	7	32
cycle direct anchors	3	24
cycle restored anchors	4	12
ordinary tree identity	3	1
total		176

There are 143 isomorphism and 33 triangle anchors.

If a rooted binary factor has k boundary leaves and r reticulations, degree counting gives $t = k + r - 2$ nonroot tree vertices and $2k + 3r - 2$ rooted arcs. Root suppression merges two arcs, leaving exactly

$$2k + 3r - 3$$

physical attachment sites. The probe input contains every site: the root-suppressed incoming segment, every other pendant arm, both incoming edges at every reticulation, and every remaining unheaded core edge. The two artificial root halves are identified only after exact semi-directed isomorphism.

Across all anchors there are 2,206 source sites and 2,206 target sites. The first-probe universe is the per-anchor Cartesian product and contains 29,964 directions. Its census is

$$27,758 \text{ quartet} + 99 \mathcal{T}_i + 1,915 \text{ isomorphism} + 192 \text{ triangle}.$$

The 2,107 equality survivors generate every two-port parent exactly once. The second-probe universe has 544,571 directions:

$$511,266 \text{ quartet} + 576 \mathcal{T}_i + 30,969 \text{ isomorphism} + 1,760 \text{ triangle}.$$

One-port restrictions identify the directed core segment containing each omitted boundary. Two-port restrictions determine the order of every pair on one segment. Since the comparisons come from a genuine word, they are transitive and recover a unique total order. The exact child transports agree on overlaps and restrict to their parents. All 33 triangle anchors transport the same inherited ordinary triangle; probes create no new triangle. This is the finite mechanism that extends the bounded theorem from rigid supports to arbitrary complete labelled subdivision words.

7 Representative exact calculations

7.1 Tree-sunlet factorization

For the three-sunlet orientation at leaf three,

$$q_{xyz} = a_x b_y c_z \{ \delta f_y d_z + (1 - \delta) f_x e_z \}, \quad x + y + z = 0.$$

With $X_h = q_{hh0}$, $Y_h = q_{h0h}$, $Z_h = q_{0hh}$, and $V = q_{CTG}$, exact substitution gives

$$V^2 X_g - X_s^2 Y_g Z_g = -a_s^2 b_s^2 a_g b_g c_g^2 f_s^2 \delta (1 - \delta) d_g e_g (1 - f_g)^2.$$

This identity is used on whole semi-directed maps. A rooted restriction type is not used as a topology oracle.

7.2 Bridge normalizer

For an unmarked degree- d component, the pair-anchor rows $(12), (13), (23), (14), \dots, (1d)$ have rank d and leading determinant -2 . If r_{ij} are positive anchor ratios, the incidence scales are

$$a_1 = \sqrt{r_{12}r_{13}/r_{23}}, \quad a_2 = \sqrt{r_{12}r_{23}/r_{13}}, \quad a_3 = \sqrt{r_{13}r_{23}/r_{12}}, \quad a_k = r_{1k}/a_1.$$

The formula is applied independently to the $\{C, T\}$ and G sectors.

7.3 Certificate schema for a direct polynomial

A sparse direct certificate stores

$$P(q) = \sum_m c_m \prod_{j \in m} q_j,$$

with integer coefficients and explicit Fourier-coordinate tuples. Its replay regenerates source and target polynomial maps from the two graph encodings, forms $P \circ \Phi_S$ and $P \circ \Phi_T$ over $\mathbb{Z}$, checks $P \circ \Phi_T = 0$ coefficientwise, checks a nonzero source polynomial hash, and evaluates it at a rational point satisfying every strict K2P and inheritance inequality. Degrees two, three, four, and five are stored and mutated separately so that certificate reassignment is rejected.

8 Printed exact-certificate appendix

This appendix exposes the formulas already derived by the first compression cycle; it is not a second compression claim. The 23 quadratic templates identify literal coefficient, coordinate-pair, and multidegree bodies only. Equality of displayed formulas across layers does not license any graph, port, direction, or coordinate transport. In particular, the repeated literal bodies in different layers remain different graph-derived certificates.

8.1 Coordinate dictionaries

Write the character group in the integer order $0, C, G, T$. A coordinate index is obtained by listing the zero-sum tuples, replacing each tuple by the lexicographically smaller of it and its global C/T swap, deleting duplicates, and sorting. Thus $q_i = q_{h_1 \dots h_k}$ for the tuple printed below. All 36 four-port coordinates occur in the printed quadratic or higher-degree formulas:

index	tuple	index	tuple	index	tuple	index	tuple
q_0	0000	q_1	00CC	q_2	00GG	q_3	0C0C
q_4	0CC0	q_5	0CGT	q_6	0CTG	q_7	0G0G
q_8	0GCT	q_9	0GG0	q_{10}	C00C	q_{11}	C0C0
q_{12}	C0GT	q_{13}	C0TG	q_{14}	CC00	q_{15}	CCCC
q_{16}	CCGG	q_{17}	CCTT	q_{18}	CG0T	q_{19}	CGCG
q_{20}	CGGC	q_{21}	CGT0	q_{22}	CT0G	q_{23}	CTCT
q_{24}	CTG0	q_{25}	CTTC	q_{26}	G00G	q_{27}	G0CT
q_{28}	G0G0	q_{29}	GC0T	q_{30}	GCCG	q_{31}	GCGC
q_{32}	GCT0	q_{33}	GG00	q_{34}	GGCC	q_{35}	GGGG

The following are exactly the 52 five-port indices occurring in the stored literal coordinate-pair blocks behind the printed theta2 and restoration quadratics. Of these, 46 have a nonzero displayed coefficient; six occur only in zero slots retained by the exact certificate-body normalization:

index	tuple	index	tuple	index	tuple	index	tuple
q_1	000CC	q_2	000GG	q_3	00C0C	q_4	00CC0
q_5	00CGT	q_6	00CTG	q_7	00G0G	q_8	00GCT
q_9	00GG0	q_{10}	0C00C	q_{18}	0CG0T	q_{26}	0G00G
q_{27}	0G0CT	q_{28}	0G0G0	q_{29}	0GC0T	q_{30}	0GCCG
q_{31}	0GCGC	q_{32}	0GCT0	q_{34}	0GGCC	q_{35}	0GGGG
q_{36}	C000C	q_{37}	C00C0	q_{40}	C0C00	q_{41}	C0CCC
q_{43}	C0CTT	q_{49}	C0TCT	q_{51}	C0TTC	q_{52}	CC000
q_{53}	CC0CC	q_{55}	CC0TT	q_{56}	CCC0C	q_{57}	CCCC0
q_{64}	CCT0T	q_{67}	CCTT0	q_{85}	CT0CT	q_{87}	CT0TC
q_{88}	CTC0T	q_{91}	CTCT0	q_{96}	CTT0C	q_{97}	CTTC0
q_{100}	G000G	q_{102}	G00G0	q_{103}	G0C0T	q_{105}	G0CGC
q_{109}	G0GGG	q_{110}	GC00T	q_{118}	GCG0C	q_{128}	GG0GG
q_{129}	GGC0C	q_{131}	GGCGT	q_{133}	GGG0G	q_{135}	GGGG0

8.2 The 23 literal quadratic bodies

The multidegree is ordered as $(s_0, g_0, s_1, g_1, \dots)$. “Canonical” counts exact certificate classes; “raw” counts their raw directional presentations.

ID	layer	polynomial	multidegree	canonical	raw
R4Q-01	four-port	$q_4q_{20} + q_5q_{21} - q_8q_{24}$ $- q_9q_{15} - q_9q_{17} + q_9q_{23}$	$(1, 0, 1, 1, 1, 1, 0)$	12	12
R4Q-02	four-port	$q_{10}q_{30} + q_{13}q_{29} - q_{15}q_{26}$ $- q_{17}q_{26} - q_{22}q_{27} + q_{23}q_{26}$	$(1, 1, 1, 0, 1, 0, 1, 1)$	52	84
R4Q-03	four-port	$q_{10}q_{20} - q_{12}q_{18}$	$(2, 0, 0, 1, 0, 1, 2, 0)$	252	424
R4Q-04	four-port	$q_3q_{19} - q_6q_{18} - q_7q_{23}$ $+ q_8q_{22}$	$(1, 0, 1, 1, 1, 0, 1, 1)$	24	40
R4Q-05	four-port	$q_3q_{19} + q_6q_{18} - q_7q_{23}$ $- q_8q_{22}$	$(1, 0, 1, 1, 1, 0, 1, 1)$	68	108
R4Q-06	four-port	$q_1q_{34} - q_8q_{27}$	$(0, 1, 0, 1, 2, 0, 2, 0)$	292	456
R4Q-07	four-port	$q_{11}q_{31} + q_{12}q_{32} - q_{23}q_{28}$ $- q_{24}q_{27}$	$(1, 1, 1, 0, 1, 1, 1, 0)$	8	8
R4Q-08	four-port	$q_1q_{16} - q_2q_{17} + q_5q_{13}$ $- q_6q_{12}$	$(1, 0, 1, 0, 1, 1, 1, 1)$	131	220
T2Q-01	five-port theta2	$q_2q_{133} - q_7q_{128} - q_{26}q_{109}$ $+ q_{35}q_{100}$	$(0, 1, 0, 1, 0, 1, 0, 1, 0, 2)$	24	60
T2Q-02	five-port theta2	$q_2q_{135} - q_9q_{128} - q_{28}q_{109}$ $+ q_{35}q_{102}$	$(0, 1, 0, 1, 0, 1, 0, 2, 0, 1)$	24	60
T2Q-03	five-port theta2	$q_{36}q_{91} + q_{37}q_{64} - q_{40}q_{87}$ $- q_{49}q_{52}$	$(2, 0, 1, 0, 1, 0, 1, 0, 1, 0)$	16	40
T2Q-04	five-port theta2	$q_{36}q_{57} - q_{36}q_{67} - q_{37}q_{56}$ $+ q_{37}q_{88} + q_{40}q_{55} - q_{40}q_{87}$ $- q_{43}q_{52} + q_{49}q_{52}$	$(2, 0, 1, 0, 1, 0, 1, 0, 1, 0)$	32	80
C3Q-01	three-port cycle in four-port coordinates	$q_4q_{20} + q_5q_{21} - q_8q_{24}$ $- q_9q_{15} - q_9q_{17} + q_9q_{23}$	$(1, 0, 1, 1, 1, 1, 1, 0)$	2	4
C3Q-02	three-port cycle in four-port coordinates	$q_4q_{30} - q_6q_{32}$	$(0, 1, 2, 0, 2, 0, 0, 1)$	16	40
C3Q-03	three-port cycle in four-port coordinates	$q_1q_{34} - q_8q_{27}$	$(0, 1, 0, 1, 2, 0, 2, 0)$	16	40
C3Q-04	three-port cycle in four-port coordinates	$q_1q_{16} - q_2q_{17} - q_5q_{13}$ $+ q_6q_{12}$	$(1, 0, 1, 0, 1, 1, 1, 1)$	2	4
C3Q-05	three-port cycle in four-port coordinates	$q_1q_{16} - q_2q_{17} + q_5q_{13}$ $- q_6q_{12}$	$(1, 0, 1, 0, 1, 1, 1, 1)$	2	4
C3Q-06	three-port cycle in four-port coordinates	$q_{11}q_{19} - q_{13}q_{21}$	$(2, 0, 0, 1, 2, 0, 0, 1)$	16	40
RSQ-01	five-port restoration	$q_3q_{131} - q_5q_{129} - q_{29}q_{105}$ $+ q_{31}q_{103}$	$(0, 1, 0, 1, 2, 0, 0, 1, 2, 0)$	1	1

ID	layer	polynomial	multidegree	canonical	raw
RSQ-02	five-port restoration	$q_1 q_{34} - q_8 q_{27}$	(0, 0, 0, 1, 0, 1, 2, 0, 2, 0)	2	6
RSQ-03	five-port restoration	$q_4 q_{30} - q_6 q_{32}$	(0, 0, 0, 1, 2, 0, 2, 0, 0, 1)	1	10
RSQ-04	five-port restoration	$q_{10} q_{118} - q_{18} q_{110}$	(0, 1, 2, 0, 0, 1, 0, 0, 2, 0)	1	130
RSQ-05	five-port restoration	$q_3 q_{131} - q_5 q_{129} + q_{29} q_{105} - q_{31} q_{103}$	(0, 1, 0, 1, 2, 0, 0, 1, 2, 0)	1	1

8.3 Five high-degree bases and their licensed transports

Only the 22 port permutations, 12 quartic coordinate transports, and two cubic record bindings printed here are licensed. No other port, source-target, inheritance, pole/sink, or C/T transport is inferred.

HDQ-01: theta0 quintic. Degree 5, multidegree (2, 1, 1, 1, 2, 1, 1, 1).

$$\begin{aligned}
F_{HDQ-01} = & q_0 q_0 q_{11} q_{17} q_{35} - q_0 q_0 q_{11} q_{25} q_{35} - 2q_0 q_0 q_{17} q_{19} q_{28} + 2q_0 q_0 q_{19} q_{25} q_{28} \\
& - 2q_0 q_1 q_{11} q_{14} q_{35} + 2q_0 q_1 q_{13} q_{24} q_{33} + 4q_0 q_1 q_{14} q_{19} q_{28} - 2q_0 q_1 q_{21} q_{24} q_{26} \\
& + 2q_0 q_2 q_{10} q_{21} q_{32} + q_0 q_2 q_{11} q_{17} q_{33} - 2q_0 q_2 q_{11} q_{21} q_{29} - 2q_0 q_2 q_{11} q_{23} q_{33} \\
& + q_0 q_2 q_{11} q_{25} q_{33} - 2q_0 q_2 q_{14} q_{21} q_{27} + 2q_0 q_4 q_{10} q_{11} q_{35} - 4q_0 q_4 q_{10} q_{19} q_{28} \\
& - 2q_0 q_4 q_{12} q_{13} q_{33} + 2q_0 q_4 q_{12} q_{21} q_{26} - 2q_0 q_5 q_{11} q_{13} q_{33} + 2q_0 q_5 q_{11} q_{21} q_{26} \\
& + q_0 q_7 q_{11} q_{17} q_{28} - q_0 q_7 q_{11} q_{25} q_{28} - 2q_0 q_9 q_{10} q_{13} q_{32} + 2q_0 q_9 q_{11} q_{13} q_{29} \\
& - q_0 q_9 q_{11} q_{17} q_{26} + 2q_0 q_9 q_{11} q_{23} q_{26} - q_0 q_9 q_{11} q_{25} q_{26} + 2q_0 q_9 q_{13} q_{14} q_{27} \\
& - 2q_1 q_7 q_{11} q_{14} q_{28} + 2q_2 q_3 q_{11} q_{11} q_{33} - 2q_3 q_9 q_{11} q_{11} q_{26} + 2q_4 q_7 q_{10} q_{11} q_{28}
\end{aligned}$$

HDQ-02: lower-theta quartic. Degree 4, multidegree (3, 0, 2, 1, 1, 1, 2, 1).

$$\begin{aligned}
F_{HDQ-02} = & q_0 q_{15} q_{16} q_{18} + q_0 q_{16} q_{18} q_{23} - q_0 q_{20} q_{22} q_{23} + q_1 q_{14} q_{16} q_{18} \\
& - 2q_3 q_{11} q_{16} q_{18} + q_3 q_{11} q_{20} q_{22} - q_4 q_{12} q_{18} q_{22} - q_8 q_{10} q_{14} q_{16} \\
& + q_8 q_{12} q_{14} q_{22}
\end{aligned}$$

HDQ-03: lower-theta quartic. Degree 4, multidegree (2, 1, 3, 0, 1, 1, 2, 1).

$$\begin{aligned}
F_{HDQ-03} = & q_0 q_{16} q_{17} q_{29} - q_0 q_{16} q_{23} q_{29} - q_0 q_{17} q_{22} q_{31} + q_0 q_{22} q_{23} q_{31} \\
& - q_1 q_{14} q_{16} q_{29} + q_1 q_{14} q_{22} q_{31} + 2q_3 q_{11} q_{16} q_{29} - q_3 q_{11} q_{22} q_{31} \\
& - q_3 q_{14} q_{16} q_{27} - q_5 q_{11} q_{22} q_{29} + q_5 q_{14} q_{22} q_{27}
\end{aligned}$$

HDQ-04: lower-theta quartic. Degree 4, multidegree (2, 1, 3, 0, 2, 1, 1, 1).

$$\begin{aligned}
F_{HDQ-04} = & q_0 q_{15} q_{16} q_{32} + q_0 q_{16} q_{23} q_{32} - q_0 q_{23} q_{24} q_{30} + q_1 q_{14} q_{16} q_{32} \\
& - 2q_3 q_{11} q_{16} q_{32} + q_3 q_{11} q_{24} q_{30} - q_4 q_{14} q_{16} q_{27} - q_6 q_{10} q_{24} q_{32} \\
& + q_6 q_{14} q_{24} q_{27}
\end{aligned}$$

HDQ-05: theta3 cubic. Degree 3, multidegree (1, 1, 1, 1, 1, 1, 1, 1).

$$\begin{aligned}
F_{HDQ-05} = & q_0 q_{15} q_{35} + q_0 q_{20} q_{30} + q_1 q_{14} q_{35} + q_2 q_{15} q_{33} \\
& + q_2 q_{18} q_{32} - q_2 q_{23} q_{33} - q_3 q_{11} q_{35} - q_4 q_{20} q_{26} \\
& + q_6 q_{12} q_{33} - q_6 q_{18} q_{28} - q_7 q_{12} q_{32} - q_7 q_{15} q_{28} \\
& + q_7 q_{23} q_{28} - q_9 q_{10} q_{30}
\end{aligned}$$

Quintic port transports.

class	port permutation	source terms
25	(0, 1, 3, 2)	1599
26	(0, 2, 1, 3)	953
27	(0, 2, 3, 1)	1524
28	(0, 3, 1, 2)	985
29	(0, 3, 2, 1)	96
30	(1, 0, 2, 3)	730
31	(1, 0, 3, 2)	1988
32	(1, 2, 0, 3)	953
33	(1, 2, 3, 0)	2004
34	(1, 3, 0, 2)	985
35	(1, 3, 2, 0)	868
36	(2, 0, 1, 3)	730
37	(2, 0, 3, 1)	1722
39	(2, 1, 3, 0)	1807
40	(2, 3, 0, 1)	96
41	(2, 3, 1, 0)	868
42	(3, 0, 1, 2)	1988
43	(3, 0, 2, 1)	1722
44	(3, 1, 0, 2)	1599
45	(3, 1, 2, 0)	1807
46	(3, 2, 0, 1)	1524
47	(3, 2, 1, 0)	2004

Quartic coordinate transports.

source	class	base	coordinate action	port match
2	112	HDQ-02	identity	(0, 1, 2, 3)
2	113	HDQ-03	identity	(0, 1, 3, 2)
2	114	HDQ-03	identity	(1, 0, 2, 3)
2	115	HDQ-02	identity	(1, 0, 3, 2)
3	112	HDQ-02	identity	(0, 1, 2, 3)
3	113	HDQ-03	identity	(0, 1, 3, 2)
3	114	HDQ-03	identity	(1, 0, 2, 3)
3	115	HDQ-02	identity	(1, 0, 3, 2)
4	8	HDQ-04	identity	(0, 1, 2, 3)
4	9	HDQ-04	swap_01	(0, 1, 3, 2)
4	10	HDQ-04	swap_01	(1, 0, 2, 3)
4	11	HDQ-04	identity	(1, 0, 3, 2)

Cubic record bindings.

source	class	base	port match
5	9	HDQ-05	(0, 1, 3, 2)
5	10	HDQ-05	(1, 0, 2, 3)

8.4 Three worked paths

A direct quadratic separator. Template R4Q-03 is

$$q_{10}q_{20} - q_{12}q_{18}, \quad \text{mdeg} = (2, 0, 0, 1, 0, 1, 2, 0).$$

It covers 252 canonical classes and 424 raw directions. For the representative `source_0:class_000000`, its source pullback has 44 nonzero terms and its target pullback is identically zero. The exact certificate is `d39c8b1f09360f48e9ae6e6ed55222d35cdc675fdd22c038781a7ff5e0328f5`, bound to the graph record by `142e31e65f4ebab77b8523207ee27ac99db0e8f5e58946f289738790ad0a8782`. This proves the stated source-to-target noncontainment; reversing the direction is not licensed.

A complete restoration root. For `s1:c658:t2792:p1032`, the following table prints every actual first- and second-child archetype. The row hash binds the role, insertion site, proof mechanism, transport fields, and evidence certificate.

layer	row	restored role	site	proof	status	row SHA-256
1	31976	D_REPAIR_O_2	0	displayed_quartet_mismatch	separated	d2a1d79aaa37de82495c0db5478793b02dc15b7500f2783506614a3afc46531d
1	31977	D_REPAIR_O_2	1	displayed_quartet_mismatch	separated	18a009d352b311939bb21650fba5d66ce0a1a8dbe29ebaa7bd6393e0c0d25748
1	31978	D_REPAIR_O_2	2	full_map_Ti_zero_strict_sign	separated	a762631aa3a30f73764549778fcd2fceb9bcaac256430648a4f4ab1bdabb4869
1	31979	D_REPAIR_O_2	3	displayed_quartet_mismatch	separated	181aad684be16b6b531b5c2a8a8cbbd5ab4492018427e921140695868c4bc29e
1	31980	D_REPAIR_O_2	4	displayed_quartet_mismatch	separated	a37508b280187028982572253c3a59826137feb537228fb443fb79b684543626
1	31981	D_REPAIR_O_2	5	displayed_quartet_mismatch	separated	d7a4960b94bf787bdc5ddeafad029506a4693b5aa5876f5ecb947500abcf1a6c
1	31982	D_REPAIR_O_2	6	displayed_quartet_mismatch	separated	d5ad83a75f35c4b1bd795d89363c7b147d10ff6629930c327b3916883cd75ba2
1	31983	INCOMING	0	displayed_quartet_mismatch	separated	8246f706cb8e56b848238ca6e8715cc7ec257691cc03f567b97cac4843ec400a
1	31984	INCOMING	1	displayed_quartet_mismatch	separated	12bba2049bad09e6403af99ae1cd8f54003aafd14654396f2328220ac954b079
1	31985	INCOMING	2	displayed_quartet_mismatch	separated	8893a4e1763de61a221d08388243e53f69e501f2e89cee01d493e6ae6eb75f95
1	31986	INCOMING	3	displayed_quartet_mismatch	separated	07b72f4608262f6fc6958a0f24c966984c5d8d70c08f490abfa216cadb2c801e
1	31987	INCOMING	4	restore_remaining_physical_role	continuation	60636f150c4eb58cd64c0126bdb1d4aafe8e48927fc798974dfba76a5c51d272
1	31988	INCOMING	5	displayed_quartet_mismatch	separated	718c7d8d3fd230a2bd1b25ecbb0faf7f59a6bb6a8b0cd261f530806b3aac25b7
1	31989	INCOMING	6	displayed_quartet_mismatch	separated	b86c1e7ab3211f299a5b9eb5910e3ef3b3ae35f69065cf189be6e31d04d288b8
2	1	D_REPAIR_O_2	0	displayed_quartet_mismatch	separated	daddc54b748d86b7f3698b771e1ea090b3fee641becc9b68fb76b0775d2b48e1

layer	row	restored role	site	proof	status	row SHA-256
2	2	D_REPAIR_O_2	1	displayed_quartet_mismatch	separated	3d678ac9082f6b02ffebbb34085488455455345b13f1e037681b75051c81a603e
2	3	D_REPAIR_O_2	2	full_map_Ti_zero_strict_sign	separated	dbe28657406837bfe91de37dc42fb124c9a71d2c475bf1655811deeeac55ed00
2	4	D_REPAIR_O_2	3	displayed_quartet_mismatch	separated	e655cf2cf0aa1572b4a83ad7b24bbe fde08e7fd968b0906c31cc3beb6977d72a
2	5	D_REPAIR_O_2	4	displayed_quartet_mismatch	separated	f61fd9ab276190e4e08a3e91288d012d46fad0e92ef586e74c9c20da566280d2
2	6	D_REPAIR_O_2	5	displayed_quartet_mismatch	separated	5145c0082ec9fc0bfc40d4315a851cf39f634e4ea3b14504f3f455a8139b0233
2	7	D_REPAIR_O_2	6	displayed_quartet_mismatch	separated	61bd0a2485ca2ba94809531aae4245dd53821a10a041f95ff1e33b1f20f118a1
2	8	D_REPAIR_O_2	7	displayed_quartet_mismatch	separated	1b825df66c3c7d4f9c9cf2156fa16cc7dd1fe9dc0849deb83d0203bfcea3a64d

There are 14 first children: 12 displayed-quartet terminals, one full-map $\mathcal{T}_i$ terminal, and one continuation (row 31987). That continuation has exactly eight children: seven displayed-quartet terminals and one full-map $\mathcal{T}_i$ terminal. Every second child has an empty remaining-role list. Hence this exact subtree has 21 leaves, depth 2, zero unresolved records, and no cycle.

A one-/two-port word transport. The anchor `tree:k3:identity` is the labelled identity tree transport `4342a65ef53537888b371724949a9ec3260731896777270eceed8fea0b475a7f`. At source and target site zero, inserting label 3 gives the exact one-port transport `c2ed6cbcd9d725664268416f315268cfdab6e2c2844790c7596aab7010e0a4e0`; both parent restrictions are `R:57a56588cb2934a8ad6d9500e0d28a8f7a3e8f797ba05e6a213af02a676eb539`. Inserting label 4 at the new site zero gives the exact two-port transport `8e3c37f3b5c32225fbf97286be01da6754a33e4fd267e4741a918740d5a4c8ba`, whose parent transport is precisely the one-port transport. Its two restriction records are `R:b286c2133804e765ea7889cbdc790218c360023201fbd93991e608af071e4bd3` and `R:0bf3e2b848168b2d5c1d689601f1c1700bd3f98502524f32763d20ddac093639`. The reversed marginal is the exact one-port transport `c144b9418ca2bc530028a613a74290e7ded74d9b38589f323cd7e01b7bdf33b6` on the same base anchor. The selected ledger rows have hashes `67691e6e518436f8911fe0dc82402de996b0deef87e3f7ad6396fb0b81bd0ef0` and `44c2b033d79916a3226985efc6211b4b54844953f90a7c00b2788a0cfe206193`. Thus both deletion orders are bound, not inferred from an unrecorded symmetry.

8.5 Provenance and replay

field	exact value
schema/version	k2p-printed-certificate-appendix-v1
producer	.env/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -write
deterministic check	.env/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -check
independent replay	.env/bin/python -B proof_compression_submission/templates/verify_printed_certificate_appendix.py
mutation suite	.env/bin/python -B proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py
sealed payload	afc1b2622943e07330b24e0195bef28d898d63f032813b9587ec6f7df175c1b2

The authoritative input bindings are:

path	SHA-256
proof_compression_submission/templates/DIRECT_CERTIFICATE_TEMPLATE_TABLE.json	b6c63c7818c54ac0bae1e4d3f64047cbcaa531f83ad0b16e6a4ec92f134f7698
work/corrected_composite_ledgers/artifacts/raw4_terminal_certificate_registry.json.gz	8d821c2000da5cf2647913cbdb42f8a42df6b6826b8b76be49d91d78ebaf9998
package/referee/k2p_offline_sweep_portable/proofs/theta0_quintic_orbit_certificate.json	f863afd5875a74be818141990863344fae09fd4269a803d3a1bf6b67a8a595e0
package/referee/k2p_offline_sweep_portable/proofs/theta_quartic_obstruction_certificates.json	1242a016b200bafd6326366d23fc85c080fd1eb6d29882d4e2567b074ab24965
package/referee/k2p_offline_sweep_portable/proofs/theta3_cubic_obstruction_certificate.json	aae0f05bcff804da4965d5a7123201b84bcb71e4f0f98d91d2defaa0fedc304c
package/referee/k2p_offline_sweep_portable/proofs/four_port_direct_residual_closure_certificate.json	8f0760543d0b69937c24785288ee26f58db86f34bb3446d686e0422eb2fa7af7
work/restoration_sign_reclassification/corrected_restoration_forest.json	396d1970af17b5e90c3f1b00ceab1b810816e93ec68a566bd0479f05c722793f
work/probe_coherence_corrected/probe_coherence_certificate.json	aef621cdefb7a892b396ae993b35d4582faa6d617a90e975d067aed2f9a53554
work/probe_coherence_corrected/one_port_ledger.jsonl.gz	d5fa13d38731bff2403eeb4e4d9e139566c4983b09d30553c6260eaac64c5c90
work/probe_coherence_corrected/two_port_ledger.jsonl.gz	10f0afcab77f2d61cecfcc36d723c6f32065c304ac088b0b8ecf12dfc867fbf9d
work/probe_coherence_corrected/parent_restriction_ledger.jsonl.gz	5d1e6c2fe38d31f6304a76886ec37829215b88c8b179f5b23596d49d37ceeb38
work/probe_coherence_corrected/exact_transport_ledger.jsonl.gz	6bc8e88feac2bee68491287775f078e8e5474bf930961a7390967c9fd350044d

9 Full weak-class certificate

The base graphs are encoded by

$$\begin{aligned}
W : \quad & rS, rL_0, SU, SV, UX, VZ, ZX, UV, ZL_1, XL_2, & \text{Ret} = \{V, X\}, \\
W' : \quad & rS, rL_0, SU, SX_0, VX_0, UX_1, VX_1, UV, X_0L_1, X_1L_2, & \text{Ret} = \{X_0, X_1\}.
\end{aligned}$$

Complete rooting censuses are (5, 2, 3) and (7, 2, 5) for admissible, tree-child, and non-tree-child rootings. Exact mixed-incidence comparison returns relation **none**, including after forgetting head flags on the unique triangle of each graph.

Put $\delta = 2^{-30}$. The complete parameter mapping is:

datum	W	W'
seven internal pairs	(1/7, 1/7)	(1/4, 1/4)
inheritance 1	$Z \rightarrow X : 15996/16339; U \rightarrow X : 343/16339$	$V \rightarrow X_1 : 1/2; U \rightarrow X_1 : 1/2$
inheritance 2	$S \rightarrow V : 1/8; U \rightarrow V : 7/8$	$V \rightarrow X_0 : 1/6; S \rightarrow X_0 : 5/6$
leaf 0 pair	(86779 δ /80, 86779 δ /80)	(16 δ /3, 16 δ /3)
leaf 1 pair	(320 δ /253, 320 δ /253)	(32 δ /9, 32 δ /9)
leaf 2 pair	(114373 δ /20240, 114373 δ /20240)	(96 δ /5, 96 δ /5)

In character code order 0, $C, G, T = 0, 1, 2, 3$, the common coordinate order is

$$(000, 0CC, 0GG, C0C, CC0, CGT, CTG, G0G, GCT, GG0).$$

The value is 1 at 000, δ^2 at the six two-nonzero coordinates, and $4\delta^3/5$ at the three all-nonzero coordinates.

The normalized tensors before pendant scaling are, respectively,

$$\left(1, \frac{64009}{457492}, \frac{64009}{457492}, \frac{6400}{39229939}, \frac{1}{1372}, \frac{4048}{39229939}, \frac{4048}{39229939}, \frac{6400}{39229939}, \frac{4048}{39229939}, \frac{1}{1372}\right)$$

and

$$\left(1, \frac{15}{1024}, \frac{15}{1024}, \frac{5}{512}, \frac{27}{512}, \frac{9}{4096}, \frac{9}{4096}, \frac{5}{512}, \frac{9}{4096}, \frac{27}{512}\right).$$

The graph-derived canonical orders of the visible edge classes are

$$W : (ZX, SV, rS, SU, UV, VZ, UX), \quad W' : (VX_1, VX_0, UV, rS, SX_0, SU, UX_1).$$

Accordingly, columns $0, \dots, 8$ mean

$$\begin{aligned} W &:: (s_{ZX}, g_{ZX}, s_{SV}, g_{SV}, s_{rS}, g_{rS}, s_{SU}, g_{SU}, s_{UV}), \\ W' &:: (s_{VX_1}, g_{VX_1}, s_{VX_0}, g_{VX_0}, s_{UV}, g_{UV}, s_{rS}, g_{rS}, s_{SX_0}). \end{aligned}$$

These names are a derived crosswalk, not an unstated field of the frozen certificate: the crosswalk verifier reconstructs the descendant signatures from the two primitive graphs, binds the frozen graph and descriptor hashes, and rejects a reordered column. The sealed artifact and independent checker are, respectively,

```
proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json
proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py.
```

All output-row indices in the following table are zero-based in the displayed Fourier-coordinate order; row 0 is the all-zero coordinate. The minors' rows and determinants are

map	output rows	determinant
W	(1, 2, 3, 5, 4, 7, 6, 8, 9)	10368019213741323/563981315074464023964442388464888915634290688
W'	(1, 2, 3, 5, 4, 6, 7, 8, 9)	1435825/85002596691653613846528

For cherry induction use actual physical pairs $(u_s, u_g) = (2/5, 4/9)$ and $(v_s, v_g) = (3/7, 5/11)$. The observable block $(u_s/v_s, u_s v_s, u_g/v_g, u_g v_g)$ has determinant $2464/675$. More explicitly, for an outside leaf j and new cherry leaves a, b ,

$$\frac{Q_{j=C, a=C, b=0}}{Q_{j=C, a=0, b=C}} = \frac{u_s}{v_s}, \quad Q_{a=C, b=C} = u_s v_s,$$

with all omitted characters zero; replacing C by G gives the other two observables. Square-root recovery gives the four edge factors, and division by the corresponding nonzero factor recovers every old tensor coordinate. This is the analytic local inverse; graph pruning separately preserves weak-not-strong status and inequivalence.

10 Theorem-to-artifact crosswalk

All paths below are relative to the `k2p_level2_identifiability_closure` project root. “Primary” is the producer/certificate path; “replay” is a separately implemented checker or audit. Historical files not named here are not promotion evidence. This printed table is a reader map, not the complete machine schema. The field-rich authority is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`, which records each claim’s declared artifact schemas, exact byte hashes, producer, replay and mutation artifacts, environment profile, and whether a runtime is known or deliberately null. Executable entry points are invoked from the project root with the pinned Python environment shown in Section 12.

claim	primary proof/certificate	independent replay or mutation
K2P domain, subdivision, rooting	work/domain_rooting_closure/PROOF.md; work/domain_rooting_closure/domain_rooting_certificate.json	work/domain_rooting_closure/verify_domain_rooting.py
Quartet signs and tree of blobs	work/quartet_separation_closure/PROOF.md; work/quartet_separation_closure/quartet_logic_certificate.json	work/quartet_separation_closure/verify_quartet_logic.py; work/adversarial_proof_review/verify_topology_direction.py
Whole-map $\mathcal{T}_i$ and triangle germ	package/original/checkpoint_2/continuation_2/K2P_TREE_SUNLET_SIGN_CERTIFICATE.md; package/original/checkpoint_2/continuation_2/K2P_TRIANGLE_GERM_EXACT.md	package/original/checkpoint_2/continuation_2/verify_triangle_and_sunlet.py; work/final_theorem_release/no_assert_triangle_sunlet.py
Bridge fibre, marginal, gluing	work/bridge_marginal_closure/PROOF.md; work/bridge_marginal_closure/certificate.json	work/bridge_marginal_closure/verify_bridge_marginal.py; work/adversarial_proof_review/PHYSICAL_LOCAL_PRODUCT_REPAIR.md; work/adversarial_proof_review/verify_adversarial.py
Global classification, genericity, reconstruction, and continuous time	proof_compression_submission/article/main.tex; work/global_theorem_closure/promotion_manuscript/K2P_SAME_PROMOTION_MANUSCRIPT.md	work/global_theorem_closure/promotion_manuscript/QUANTIFIER_AUDIT.md; work/global_proof_adversary/AUDIT.md
Four-port raw universe and terminals	work/corrected_composite_ledgers/artifacts/raw4_corrected_composite_ledger.jsonl.gz; work/corrected_composite_ledgers/artifacts/raw4_terminal_certificate_registry.json.gz	work/corrected_composite_ledgers/verify_corrected_composites_independent.py; work/corrected_composite_ledgers/run_composite_mutations.py
Five-port θ_2	work/corrected_composite_ledgers/artifacts/theta2_corrected_composite_ledger.jsonl.gz	work/corrected_composite_ledgers/verify_corrected_composites_independent.py; work/corrected_composite_ledgers/run_composite_mutations.py, family theta2

claim	primary proof/certificate	independent replay or mutation
Production direct residual closure	package/referee/k2p_offline_sweep_portable/proofs/four_port_direct_residual_closure_certificate.json; package/referee/k2p_offline_sweep_portable/DIRECT_CLOSURE_LOCK.json	package/referee/k2p_offline_sweep_portable/verify_direct_closure_release.py; package/referee/k2p_offline_sweep_portable/test_direct_closure_release_mutations.py; package/referee/k2p_offline_sweep_portable/direct_closure_mutation_report.json
Printed direct templates and worked paths	proof_compression_submission/templates/PRINTED_CERTIFICATE_APPENDIX.json; proof_compression_submission/supplement/certificate_appendix.tex	proof_compression_submission/templates/verify_printed_certificate_appendix.py; proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py (reader-facing presentation regression only; theorem authority remains the production direct-closure replay and mutation suite)
997-parent forest	work/restoration_sign_reclassification/corrected_restoration_forest.json	work/restoration_sign_reclassification/corrected_restoration_replay_certificate.json; work/restoration_sign_reclassification/verify_corrected_restoration_forest.py; work/restoration_sign_reclassification/mutate_corrected_restoration_forest.py
Three-port cycle closure	work/cycle_three_port_closure/PROOF.md; work/cycle_three_port_closure/promotion/cycle_promotion_certificate.json	work/adversarial_proof_review/verify_corrected_cycle_promotion.py; work/adversarial_proof_review/test_cycle_promotion_mutations.py
Probe input completeness	work/adversarial_proof_review/PROBE_INPUT_CONTRACT.md; work/adversarial_proof_review/probe_input_contract.json	work/adversarial_proof_review/verify_probe_input_contract.py; work/adversarial_proof_review/test_probe_input_mutations.py

claim	primary proof/certificate	independent replay or mutation
One-/two-port probe closure	work/probe_coherence_corrected/PROOF.md; work/probe_coherence_corrected/probe_coherence_certificate.json	work/probe_coherence_corrected/probe_coherence_independent_verification.json; work/probe_coherence_corrected/verify_probe_coherence_corrected.py; work/probe_coherence_corrected/run_probe_coherence_mutations.py
Primitive probe audit	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit_certificate.json	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit.py; work/global_proof_adversary/probe_full_audit/independent_probe_mutation_report.json
Weak sharpness	work/weak_sharpness_closure/PROOF.md; work/weak_sharpness_closure/weak_sharpness_certificate.json; proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json	work/weak_sharpness_audit/PROOF_AUDIT.md; work/weak_sharpness_audit/audit_certificate.json; work/weak_sharpness_audit/test_mutations.py; work/weak_sharpness_audit/mutation_report.json; proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py; proof_compression_submission/analysis/test_weak_sharpness_column_crosswalk_mutations.py
Unified theorem lock	work/final_theorem_release/RELEASE_LOCK.json; work/final_theorem_release/verify_final_theorem_release.py	work/final_theorem_release/corrected_universe_independent_replay.json; work/final_theorem_release/run_release_mutations.py

10.1 Printed finite-layer replay metadata

For the principal computed layers, each row below supplies the requested schema/version, authoritative file SHA-256, producer command, independent replay command, and mutation command. Commands run from the project root; none may be invoked with optimized Python.

layer	exact machine metadata
raw four-port topology ledger	schema/version: k2p-four-port-raw-ledger-summary-v1; authority: work/raw_ledger_audit/artifacts/raw_ledger_summary.json; SHA-256: 59212034aca49b29d2e0dd0e312cd5e7e76957b48e8eb312f74ce277e0843f43; producer command: .venv/bin/python -B work/raw_ledger_audit/generate_raw_ledger.py; replay command: .venv/bin/python -B work/raw_ledger_audit/verify_raw_ledger.py; historical mutation-regression command (retained for provenance, not current theorem qualification): .venv/bin/python -B work/raw_ledger_audit/test_mutations.py. The current semantic rank attack is the exact-rank production-verifier suite in the next row.
four-port exact-rank filter	schema/version: k2p-four-port-exact-generic-rank-upper-coverage-v1; authority: work/rank_upper_certificates/rank_upper_coverage.json; SHA-256: c52c5730494eb894360c17b6e54ae5c260fca3cddb8702d5c796750c7df874bc; producer command: PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/build_rank_upper_coverage.py; replay command: PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/verify_rank_upper_certificates.py; mutation command: PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/mutation_tests.py -output /tmp/k2p-rank-upper-mutations.json.
direct terminal templates	schema/version: k2p-printed-certificate-appendix-v1; reader derivative: proof_compression_submission/templates/PRINTED_CERTIFICATE_APPENDIX.json; SHA-256: a06dc d55d2f834007cd5933522f0ea3b697c983758645a07311c980acb94c2ab; producer command: .venv/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -write; replay command: .venv/bin/python -B proof_compression_submission/templates/verify_printed_certificate_appendix.py; presentation-regression command (not theorem qualification): .venv/bin/python -B proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py.
997-parent restoration forest	schema/version: k2p-corrected-restoration-forest-v3; authority: work/restoration_sign_reclassification/corrected_restoration_forest.json; SHA-256: 396d1 970af17b5e90c3f1b00ceab1b810816e93ec68a566bd0479f05c722793f; producer command: .venv/bin/python -B work/restoration_sign_reclassification/build_corrected_restoration_forest.py; replay command: .venv/bin/python -B work/restoration_sign_reclassification/verify_corrected_restoration_forest.py; mutation command: .venv/bin/python -B work/restoration_sign_reclassification/mutate_corrected_restoration_forest.py -output /tmp/k2p-restoration-mutations.json.
one-/two-port probe closure	schema/version: k2p-corrected-coherent-probe-closure-v1; authority: work/probe_coherence_corrected/probe_coherence_certificate.json; SHA-256: aef621cdeffb7a8 92b396ae993b35d4582faa6d617a90e975d067aed2f9a53554; producer command: .venv/bin/python -B work/probe_coherence_corrected/build_probe_coherence_corrected.py; replay command: .venv/bin/python -B work/probe_coherence_corrected/verify_probe_coherence_corrected.py; mutation command: .venv/bin/python -B work/probe_coherence_corrected/run_probe_coherence_mutations.py -output /tmp/k2p-probe-mutations.json.
weak-sharpness named columns	schema/version: k2p-weak-sharpness-column-crosswalk-v1; authority: proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json; SHA-256: a66 29eba036b93170d27cbb72ba04cd30b9b8c0b221f81ec4f450ca9ee6eb058; producer command: .venv/bin/python -B proof_compression_submission/analysis/build_weak_sharpness_column_crosswalk.py; replay command: .venv/bin/python -B proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py; mutation command: .venv/bin/python -B proof_compression_submission/analysis/test_weak_sharpness_column_crosswalk_mutations.py.
full-map certificate reseal	schema/version: k2p-full-map-domain-reseal-audit-v1; authority: work/final_theorem_release/full_map_reseal_audit.json; SHA-256: b1eda458ede322a272f0306a 58bb87def243295ba52abc52361bcea4a5ea7fcc; producer/replay/mutation command: .venv/bin/python -B work/final_theorem_release/verify_full_map_reseal.py -output /tmp/k2p-full-map-reseal-audit.json.
composite reseal differential	schema/version: k2p-composite-domain-reseal-diff-audit-v1; authority: work/final_theorem_release/composite_reseal_diff_audit.json; SHA-256: a3dcb06f44f4085f38 6ca37bfd29809536b79d10c2c6e920b28144f97a894a68; producer/replay command: .venv/bin/python -B work/final_theorem_release/verify_composite_reseal_diff.py; mutation authority: the six fail-closed seal/domain mutations in the preceding reseal audit.

11 Frozen hash anchors

The following SHA-256 values are the compact anchors most useful for checking that the intended finite universe has been unpacked. The outer release lock authenticates all nested files and should remain the authority if any table is updated.

artifact	file SHA-256
unified release lock	528e999243f9c43bf7ac4102607f0024610fcffd71cce66eeb50ca054dbc2970
raw-four composite ledger	7cf3f953fca695d612387143818843650498f84f55cf0a776f90c9afdd95eef6
raw-four independent replay	610748d793db9c9045d0eec416f4ea97c54bc4fa4467fe69029fb06fa7db1602
raw-four terminal registry	8d821c2000da5cf2647913cbdb42f8a42dfefeb6826b8b76be49d91d78ebaf9998
raw-four mutation report	6475e37d2ec15eafc3650f387d8d4543268963c6794e3dfd90376d01741f23db
five-port θ_2 composite ledger	805fc7f5a3de9dad2c63a210208075cf19910cf811ffd08878f32782ce71b659
five-port independent replay	cd503f97e4fc0aacb52e04fa02358c95cca099d1b0117e7b7133b0913814d2cd
five-port mutation report	07f51fe0d5a28cd673cf782478c4a024e38ff7c729cf7a1e5cdafa4b53a4fac5
full-map certificate reseal audit	b1eda458ede322a272f0306a58bb87def243295ba52abc52361bcea4a5ea7fcc
composite reseal differential audit	a3dcb06f44f4085f386ca37bfd29809536b79d10c2c6e920b28144f97a894a68
corrected 997-parent forest	396d1970af17b5e90c3f1b00ceab1b810816e93ec68a566bd0479f05c722793f
restoration independent replay	d74cc01341f405732c6ff62558ca3afff705c15cdf9a6f16dcc6ccd7636749c4
probe input contract	5e6e955db206a0c2b5f520a67fd2fededcedcdef88d466a7c8e436108a77fac24
full probe primary	aef621cdefb7a892b396ae993b35d4582faa6d617a90e975d067aed2f9a53554
full probe independent graph audit	68d323419270209ee795e0b98b52045cdaf120f21046c4a232207fd838443319
full probe adversarial mutation report	d75c348e905d4d4749e9b6835abb25cb42aac24d10a0c91b5ecb64443fe07a7a
weak-sharpness primary	e66c78a0aeab990b4dc448f4f064b37e1e15ecbfff75a5f472bf116d4464378bd
weak-sharpness independent audit	cf8d83a2ebc7431d141cac6ebe943e25730eb086fbc84b52833a40bee40a5d52

The combined one-/two-port ordered-ledger root is

7868fed6f8e0c10fcb9740da8ffdcdb7f64ea68939c99cba6f364da4cfd90bf50.

12 Replay protocol

Run from the project root. The locked environment uses Python 3.10 or newer; the qualified release used Python 3.14.6, NetworkX 3.5, and SymPy 1.14.0.

```
python3 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install \
  -r work/final_theorem_release/requirements.txt

.venv/bin/python -B \
  work/final_theorem_release/build_release_lock.py --check --require-ready
.venv/bin/python -B \
  work/final_theorem_release/verify_final_theorem_release.py --quick
.venv/bin/python -B \
  work/final_theorem_release/run_release_mutations.py
```

For full primitive regeneration:

```
.venv/bin/python -B \
work/final_theorem_release/verify_final_theorem_release.py --full
```

The entry points reject optimized Python (`python -O`). Full mode regenerates large ledgers by streaming; it does not require an uncompressed multi-gigabyte temporary JSONL file. The submission bundle requires a detached clean-checkout full replay of the exact candidate sources with every layer passing, zero blockers, and `promotion_ready=true`. Its exact candidate commit, layer count, internal and wall times, memory measurements, report SHA-256, and telemetry SHA-256 are deliberately not compiled into this source. They are recorded in `proof_compression_submission/output/FINAL_CLEAN_FULL_REPLAY.json`, `proof_compression_submission/output/FINAL_CLEAN_FULL_REPLAY_TELEMETRY.json`, and the generated theorem-to-artifact crosswalk. This separation lets the replay byte-bind these exact submission sources without a self-referential source edit after the run.

The frozen five-port θ_2 artifacts retain the compiler, canonicalizer, and input-lock hashes of their original generation. The full replay requires those legacy bindings exactly and reconstructs the current expected bytes by changing only those provenance fields, the two consequent artifact-metadata rows, and the summary seal. It then compares the freshly generated rank and restoration payloads with those reconstructed bytes and requires the direct-proof, class-partition, and all 2,946,240 raw-ledger rows to remain literally byte-identical. Dedicated mutations reject a wrong legacy binding; the outer release lock independently fixes the current compiler, canonicalizer, and input lock.

13 Mutation coverage

The independent suites reject at least the following semantic faults:

- omitted or duplicated raw relation;
- swapped or false topology reason;
- false target rank upper bound or rank exclusion;
- reassignment of a terminal class;
- omitted restoration role or child;
- wrong source insertion, target attachment, or parent;
- restoration cycle or missing continuation layer;
- broken boundary transport or parent restriction;
- omitted probe anchor, pendant site, reticulation-incoming site, or root-suppressed site;
- splitting the two artificial root halves into distinct physical sites;
- source–target reversal omission;
- reassignment of quadratic, cubic, quartic, quintic, or $\mathcal{T}_i$ certificate;
- reintroduction of a rooted `triple_type` topology oracle;
- changed sharpness arrows, parents, inheritance complements, Fourier coordinates, rank minors, or incomplete rooting list; and
- optimized-Python execution that would disable fail-closed checks.

Mutation rejection demonstrates that the verifier watches these data. It is not a substitute for the direct mathematical implication associated with each certificate type.

14 Recent-source verification note

Citation metadata were initially checked on 21 August 2026 against primary publisher or authoritative preprint records where available. The entries for Brits et al. and Kriebel were rechecked against their updated primary records on 27 August 2026:

source	verified locator and status
Kimura (1980) [10]	Springer version of record, https://doi.org/10.1007/BF01731581 , vol. 16, pp. 111–120.
Evans–Speed (1993) [5]	Project Euclid/Annals record, https://doi.org/10.1214/aos/1176349030 , vol. 21, pp. 355–377.
Sturmfels–Sullivant (2005) [12]	author/arXiv and journal metadata agree on J. Comput. Biol. 12(2), pp. 204–228, https://doi.org/10.1089/cmb.2005.12.204 .
Gross et al. (2021) [6]	Springer version of record, https://doi.org/10.1007/s00285-021-01653-8 , J. Math. Biol. 83:32.
Cox–Gross–Martin (2025) [3]	Springer version of record, https://doi.org/10.1007/s11538-025-01506-1 , Bull. Math. Biol. 87:132.
Ardiyansyah (2021) [1]	arXiv primary record https://arxiv.org/abs/2104.12479v1 ; the abstract states the simple/semisimple level-2 invariant scope used in the related-work summary.
Holtgrete et al. (2025) [7]	Springer version of record, https://doi.org/10.1007/s11538-025-01549-4 , Bull. Math. Biol. 87:168; the publisher records a 14 January 2026 correction.
Holtgrete et al. (2026) [8]	Springer version of record, https://doi.org/10.1007/s12064-025-00453-8 , Theory Biosci. 145:4.
Huber et al. (2025) [9]	Springer version of record, https://doi.org/10.1007/s11538-025-01510-5 , Bull. Math. Biol. 87:136; Lemma 4.2 is the two-generator result used here.
Englander et al. (2025) [4]	bioRxiv DOI https://doi.org/10.1101/2025.04.18.649493 ; theorem numbering and the 4 July 2026 version-4 content were checked against the locally archived source XML/PDF.
Brits et al. (2026) [2]	arXiv primary record https://arxiv.org/abs/2607.12919v3 ; version 3 dated 25 August 2026. The level-1 JC/K2P/K3P result used here remains Theorem 4.9; the version-3 narrowing of the separate arbitrary-level tree–network statement to JC is not used.
Kriebel (2026) [11]	version 1.1.7 companion JC preprint, https://doi.org/10.5281/zenodo.22089373 ; its certificate and reproducibility dataset is archived separately at https://doi.org/10.5281/zenodo.22064121 .

The local K2P/K3P theta-collision manuscript is provisional and has no verified public bibliographic status in this draft. It is therefore not cited as a published result or used as theorem evidence.

15 Submission metadata and licenses

The corresponding address is me@aleckriebel.com. No specific funding supported this work, and the author declares no competing interests. The article, supplement, and certificate data are licensed under CC BY 4.0; verifier code is licensed under the MIT License. The public repository is <https://github.com/AlecKriebel/Math>, the project path is `k2p_level2_identifiability_closure`, and

the designated versioned annotated source tag for this submission is `k2p-same-biorxiv-v1.0.4`. Its tag-object and peeled-commit identifiers are external release metadata rather than self-referential source fields. The final deterministic archive SHA-256 is recorded alongside the submission package. No GitHub Release, Zenodo deposit, or DOI is claimed in this version.

Disclosure

Generative AI assisted with discovery, code, drafting, and adversarial review. The cited independent replays are separately implemented programs, not independent human review. The author accepts responsibility for the final mathematics, software, and submission.

References

- [1] Muhammad Ardiyansyah. Distinguishing level-2 phylogenetic networks using phylogenetic invariants, 2021.
- [2] Jari Brits, Niels Holtgreffe, Leo van Iersel, and Samuel Martin. On tree-network distinguishability and full identifiability of phylogenetic networks, 2026. Version 3, 25 August 2026; Theorem 4.9.
- [3] Shelby Cox, Elizabeth Gross, and Samuel Martin. Group-based phylogenetic models on 3-sunlet networks. *Bulletin of Mathematical Biology*, 87:132, 2025.
- [4] Aviva K. Englander, Martin Frohn, Elizabeth Gross, Niels Holtgreffe, Leo van Iersel, Mark Jones, and Seth Sullivant. Identifiability of phylogenetic level-2 networks under the jukes-cantor model. *bioRxiv*, 2025. bioRxiv preprint, first posted 24 April 2025; version 4 revised 4 July 2026.
- [5] Steven N. Evans and Terry P. Speed. Invariants of some probability models used in phylogenetic inference. *The Annals of Statistics*, 21(1):355–377, 1993.
- [6] Elizabeth Gross, Leo van Iersel, Remie Janssen, Mark Jones, Colby Long, and Yukihiro Murakami. Distinguishing level-1 phylogenetic networks on the basis of data generated by markov processes. *Journal of Mathematical Biology*, 83(3):32, 2021.
- [7] Niels Holtgreffe, Elizabeth S. Allman, Hector Baños, Leo van Iersel, Vincent Moulton, John A. Rhodes, and Kristina Wicke. Distinguishing phylogenetic level-2 networks with quartets and inter-taxon quartet distances. *Bulletin of Mathematical Biology*, 87(12):168, 2025.
- [8] Niels Holtgreffe, Katharina T. Huber, Leo van Iersel, Mark Jones, and Vincent Moulton. Characterizing semi-directed phylogenetic networks and their multi-rootable variants. *Theory in Biosciences*, 145:4, 2026.
- [9] Katharina T. Huber, Leo van Iersel, Mark Jones, Vincent Moulton, and Leonie Veenema-Nipius. When are quarnets sufficient to reconstruct semi-directed phylogenetic networks? *Bulletin of Mathematical Biology*, 87(10):136, 2025.
- [10] Motoo Kimura. A simple method for estimating evolutionary rates of base substitutions through comparative studies of nucleotide sequences. *Journal of Molecular Evolution*, 16(2):111–120, 1980.
- [11] Alec Kriebel. Strong tree-childness is a sharp generic-identifiability boundary for level-2 jukes-cantor networks. Zenodo preprint, 2026. Version 1.1.7; certificate and reproducibility dataset DOI: 10.5281/zenodo.22064121.
- [12] Bernd Sturmfels and Seth Sullivant. Toric ideals of phylogenetic invariants. *Journal of Computational Biology*, 12(2):204–228, 2005.